

Complete Asymptotic Transseries for the Double Factorial and Its Inverses

Bernoulli–Bell coefficient formulae, Lambert- W reversion, parity branches, and the periodic OEIS interpolation

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Abstract

The double factorial is an interleaving of two gamma-type sequences. That apparently elementary parity split is decisive for inversion: there is no canonical real inverse of a discrete sequence until an interpolation is fixed. This article first treats the two exact parity branches $\mathcal{D}_\delta(x) = (2/\pi)^{\delta/2} 2^{x/2} \Gamma(x/2 + 1)$, $\delta \in \{0, 1\}$, and then the smooth cosine interpolation recorded in OEIS A006882. With the optimal centre $X = x + 1$, both parity branches have the same logarithmic Stirling tail. Its coefficients are

$$c_k = \frac{(1 - 2^{2k-1})B_{2k}}{2k(2k-1)},$$

and all multiplicative coefficients are complete exponential Bell polynomials. For the large inverse, the dominant equation is solved exactly by a Lambert- W core. The remaining displacement contains only $T^{-1}, T^{-3}, \dots$; its coefficient functions are rational polynomials in Λ^{-1} , generated to arbitrary order by a triangular coefficient recurrence or, explicitly, by partial Bell polynomials. The first five blocks are displayed and a scaling identity connects them to the half-shifted inverse Gamma coefficients. Expanding the Lambert core gives the full outer iterated-logarithmic layer with general unsigned-Stirling-number coefficients. For the cosine interpolation, a bounded periodic factor creates an additional oscillatory $1/\Lambda$ -sector; Lagrange–Bürmann inversion gives its carrier in closed Fourier form, and a second triangular recurrence gives all algebraic corrections. Fixed-order remainders and numerical residual checks are included throughout.

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1 Introduction and scope

For integers $n \geq 0$, the double factorial sequence is determined by

$$a_n = n!!, \quad a_0 = a_1 = 1, \quad a_n = n a_{n-2}. \quad (1)$$

It is OEIS A006882 [1]. The leading estimate

$$n!! \sim \begin{cases} \sqrt{\pi} n^{(n+1)/2} e^{-n/2}, & n \text{ even}, \\ \sqrt{2} n^{(n+1)/2} e^{-n/2}, & n \text{ odd}, \end{cases}$$

is classical, but it conceals several structures needed for a genuinely all-orders inverse:

- the centre $X = n + 1$, rather than n , removes all even powers from the logarithmic correction;
- parity affects only one normalization constant after this centring;
- inversion of $X(\log X - 1)/2$ is naturally expressed by W_0 ;
- the inverse Stirling corrections live on the sparse grid $T^{-1}, T^{-3}, T^{-5}, \dots$;

- a smooth interpolation that merges both parities necessarily introduces extra information, and its inverse need not have the same transseries as either parity branch.

The coefficient calculus used below follows the Bell-polynomial and formal reversion framework developed in the ProveIt coefficient-calculus monograph [9] and the Lambert-core inversion framework of the companion series-and-transseries notes [10, 11].

Important distinction

In this article, “the inverse double factorial” means a *compositional* large real inverse of a specified analytic interpolation. The two parity branches give two canonical inverses once parity is fixed. The OEIS cosine interpolation gives a third, eventually monotone inverse with a periodic transseries sector. The reciprocal $1/(x!!)$ is different; its all-orders coefficients are also recorded in [Section 4.3](#).

The word *complete* refers to all orders of the positive-real algebraic–logarithmic transseries, with general coefficient formulae and fixed-order analytic remainders. The Stirling series is divergent. Its terminant or resurgent completion is a separate hyperasymptotic layer; the transport rule for such beyond-all-orders terms is given in [Section 10](#).

2 Exact continuations and the parity issue

2.1 The two exact parity branches

Let $\delta \in \{0, 1\}$, where $\delta = 0$ denotes the even subsequence and $\delta = 1$ the odd subsequence. Define

$$\kappa_\delta = \left(\frac{2}{\pi}\right)^{\delta/2}, \quad \mathcal{D}_\delta(x) = \kappa_\delta 2^{x/2} \Gamma\left(\frac{x}{2} + 1\right). \quad (2)$$

For every nonnegative integer $n \equiv \delta \pmod{2}$,

$$\mathcal{D}_\delta(n) = n!! . \quad (3)$$

Indeed, $(2m)!! = 2^m m!$, while $(2m-1)!! = 2^m \Gamma(m+1/2)/\sqrt{\pi}$. Equivalently, for integer n ,

$$n!! = 2^{n/2} \Gamma\left(\frac{n}{2} + 1\right) \left(\frac{2}{\pi}\right)^{(1-(-1)^n)/4}. \quad (4)$$

The normalization used repeatedly below is

$$C_\delta = \sqrt{\pi} \kappa_\delta = \begin{cases} \sqrt{\pi}, & \delta = 0, \\ \sqrt{2}, & \delta = 1. \end{cases} \quad (5)$$

2.2 The cosine interpolation recorded by OEIS

A smooth interpolation that agrees with both parity classes is

$$\mathcal{D}_c(x) = 2^{x/2} \Gamma\left(\frac{x}{2} + 1\right) \left(\frac{2}{\pi}\right)^{(1-\cos \pi x)/4}. \quad (6)$$

This is equivalent to the OEIS formula with $h = x/2$ and exponent $\sin^2(\pi h)/2$ [1]. It is only one possible interpolation, but it is particularly natural because the parity selector is an entire periodic function.

The logarithmic derivative is

$$\frac{\mathcal{D}'_c(x)}{\mathcal{D}_c(x)} = \frac{1}{2} \log 2 + \frac{1}{2} \psi\left(\frac{x}{2} + 1\right) + \frac{\pi}{4} \log\left(\frac{2}{\pi}\right) \sin \pi x. \quad (7)$$

Since $\psi(x/2 + 1) = \log(x/2) + O(x^{-1})$, this derivative is positive for all sufficiently large x . Hence $\mathcal{D}_c$ has a unique large real inverse, although that inverse differs asymptotically from both parity-resolved inverses.

2.3 Why the centre $X = x + 1$ is optimal

Put

$$X = x + 1, \quad t = \frac{X}{2}. \quad (8)$$

Then

$$\mathcal{D}_\delta(x) = \kappa_\delta 2^{t-1/2} \Gamma(t + 1/2). \quad (9)$$

The half-shift in $\Gamma(t + 1/2)$ annihilates every even inverse power in its logarithmic Stirling series. This sparse support is the reason the inverse contains only odd inverse powers of its Lambert core.

3 Coefficient toolkit

3.1 Bernoulli numbers and the half-shift identity

The Bernoulli polynomials are defined by

$$\frac{ue^{zu}}{e^u - 1} = \sum_{m \geq 0} B_m(z) \frac{u^m}{m!}, \quad B_m = B_m(0). \quad (10)$$

The value at the half-shift is

$$B_m\left(\frac{1}{2}\right) = (2^{1-m} - 1)B_m. \quad (11)$$

For odd $m > 1$, both sides vanish.

3.2 Exponential Bell polynomials

For $0 \leq m \leq n$, the partial exponential Bell polynomial is

$$B_{n,m}(x_1, x_2, \dots) = n! \sum_{\substack{j_1+j_2+\dots=m \\ j_1+2j_2+\dots=n}} \prod_{r \geq 1} \frac{1}{j_r!} \left(\frac{x_r}{r!}\right)^{j_r}. \quad (12)$$

The complete polynomial is

$$Y_n(x_1, \dots, x_n) = \sum_{m=0}^n B_{n,m}(x_1, \dots), \quad Y_0 = 1. \quad (13)$$

The two generating identities needed below are

$$\exp\left(\sum_{r \geq 1} x_r \frac{z^r}{r!}\right) = \sum_{n \geq 0} Y_n(x_1, \dots, x_n) \frac{z^n}{n!}, \quad (14)$$

$$\left(\sum_{r \geq 1} u_r z^r\right)^m = \sum_{n \geq m} \frac{m!}{n!} B_{n,m}(1!u_1, 2!u_2, \dots) z^n. \quad (15)$$

These formulae make every coefficient below a finite sum indexed by integer partitions of the target order.

3.3 Unsigned Stirling numbers of the first kind

Write $\begin{bmatrix} n \\ k \end{bmatrix}$ for the unsigned Stirling number of the first kind, defined by

$$u(u+1) \cdots (u+n-1) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} u^k. \quad (16)$$

They give the general polynomials in the large-argument expansion of Lambert W ; see [Section 6](#).

4 The complete forward transseries

4.1 Centered logarithmic form

Define the dominant phase

$$\Phi(X) = \frac{X}{2}(\log X - 1). \quad (17)$$

Theorem 4.1 (Centered double-factorial Stirling expansion). *For either $\delta \in \{0, 1\}$, as $x \rightarrow +\infty$ with $X = x + 1$,*

$$\log \mathcal{D}_\delta(x) \sim \log C_\delta + \Phi(X) + \sum_{k \geq 1} \frac{c_k}{X^{2k-1}}, \quad (18)$$

where

$$c_k = \frac{2^{2k-1} B_{2k}(1/2)}{2k(2k-1)} = \frac{(1 - 2^{2k-1}) B_{2k}}{2k(2k-1)}. \quad (19)$$

For every fixed $N \geq 0$, truncation after $c_N X^{-(2N-1)}$ has remainder

$$\log \mathcal{D}_\delta(x) - \log C_\delta - \Phi(X) - \sum_{k=1}^N \frac{c_k}{X^{2k-1}} = O\left(X^{-(2N+1)}\right). \quad (20)$$

Here and below an empty sum is understood when $N = 0$.

Proof. The duplication formula gives

$$\log \Gamma\left(t + \frac{1}{2}\right) = \log \Gamma(2t) - \log \Gamma(t) + (1 - 2t) \log 2 + \frac{1}{2} \log \pi. \quad (21)$$

Apply the ordinary positive-real Stirling expansion, with its fixed-order remainder, to the two gamma terms. Their Bernoulli corrections differ by

$$\frac{B_{2k}}{2k(2k-1)} ((2t)^{-(2k-1)} - t^{-(2k-1)}) = \frac{B_{2k}(1/2)}{2k(2k-1)t^{2k-1}}.$$

Using $t = X/2$ and multiplying by the factor $\kappa_\delta 2^{t-1/2}$ in [Equation \(9\)](#) gives [Equations \(18\)](#) and [\(19\)](#). The same subtraction of the two fixed-order Stirling remainders proves [Equation \(20\)](#); see also the standard Gamma remainder theory in [\[5, 7, 6\]](#). $\square$

The first logarithmic coefficients are

$$\begin{aligned} c_1 &= -\frac{1}{12}, & c_2 &= \frac{7}{360}, & c_3 &= -\frac{31}{1260}, & c_4 &= \frac{127}{1680}, \\ c_5 &= -\frac{511}{1188}, & c_6 &= \frac{1414477}{360360}, & c_7 &= -\frac{8191}{156}. \end{aligned} \quad (22)$$

Thus the compact all-orders form is

$$\mathcal{D}_\delta(x) \sim C_\delta \left(\frac{X}{e}\right)^{X/2} \exp\left(\sum_{k \geq 1} c_k X^{1-2k}\right). \quad (23)$$

All parity dependence has been compressed into C_δ .

4.2 Unified form for the integer sequence

For integer n , put $X = n + 1$ and $\chi_n = (-1)^n$. Then

$$C_n = \pi^{(1+\chi_n)/4} 2^{(1-\chi_n)/4} = (2\pi)^{1/4} \left(\frac{\pi}{2}\right)^{\chi_n/4}. \quad (24)$$

Consequently,

$$n!! \sim C_n \left(\frac{n+1}{e}\right)^{(n+1)/2} \exp\left(\sum_{k \geq 1} \frac{c_k}{(n+1)^{2k-1}}\right). \quad (25)$$

This is a single parity-marked asymptotic formula for OEIS A006882.

For comparison, expanding around x instead of $x + 1$ gives the denser identity

$$\log \mathcal{D}_\delta(x) \sim \frac{x}{2}(\log x - 1) + \frac{1}{2} \log x + \log C_\delta + \sum_{k \geq 1} \frac{2^{2k-1} B_{2k}}{2k(2k-1)x^{2k-1}}. \quad (26)$$

Although the logarithmic tail still has odd powers, the extra $\frac{1}{2} \log x$ complicates inversion. The shift $X = x + 1$ absorbs it into the dominant phase exactly.

4.3 Multiplicative, reciprocal, and arbitrary-power coefficients

Define the sparse sequence

$$\lambda_r = \begin{cases} c_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}. \end{cases} \quad (27)$$

For an arbitrary scalar α , set

$$\exp\left(\alpha \sum_{r \geq 1} \lambda_r X^{-r}\right) = \sum_{n \geq 0} A_n^{(\alpha)} X^{-n}. \quad (28)$$

Proposition 4.2 (General multiplicative coefficients). *For every $n \geq 0$,*

$$A_n^{(\alpha)} = \frac{1}{n!} Y_n(\alpha 1! \lambda_1, \alpha 2! \lambda_2, \dots, \alpha n! \lambda_n) \quad (29)$$

$$= \sum_{j_1+2j_2+\dots=n} \prod_{r \geq 1} \frac{(\alpha \lambda_r)^{j_r}}{j_r!}. \quad (30)$$

Equivalently,

$$A_0^{(\alpha)} = 1, \quad n A_n^{(\alpha)} = \alpha \sum_{r=1}^n r \lambda_r A_{n-r}^{(\alpha)}. \quad (31)$$

Proof. Equation (29) is the complete Bell generating identity Equation (14); expanding the Bell polynomial gives Equation (30). Logarithmic differentiation of Equation (28) and coefficient comparison gives Equation (31). $\square$

Taking $\alpha = 1$ gives

$$\mathcal{D}_\delta(x) \sim C_\delta \left(\frac{X}{e}\right)^{X/2} \sum_{n \geq 0} \frac{A_n^{(1)}}{X^n}, \quad (32)$$

whereas $\alpha = -1$ gives the reciprocal transseries

$$\frac{1}{\mathcal{D}_\delta(x)} \sim C_\delta^{-1} \left(\frac{e}{X}\right)^{X/2} \sum_{n \geq 0} \frac{A_n^{(-1)}}{X^n}. \quad (33)$$

More generally, α gives every fixed power $\mathcal{D}_\delta(x)^\alpha$ after multiplying by $C_\delta^\alpha(X/e)^{\alpha X/2}$.

The first forward coefficients are

$$\begin{array}{c|cccccc} n & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline A_n^{(1)} & 1 & -\frac{1}{12} & \frac{1}{288} & \frac{1003}{51840} & -\frac{4027}{2488320} & -\frac{5128423}{209018880} & \frac{168359651}{75246796800} \end{array} \quad (34)$$

Because the logarithmic tail is odd under $X^{-1} \mapsto -X^{-1}$, one in fact has the exact coefficient symmetry

$$A_n^{(-\alpha)} = (-1)^n A_n^{(\alpha)} \quad (n \geq 0). \quad (35)$$

In particular, the reciprocal coefficients are $A_n^{(-1)} = (-1)^n A_n^{(1)}$ at every order.

4.4 The cosine interpolation in forward form

Let

$$C_* = (2\pi)^{1/4}, \quad \rho = \frac{1}{2} \log\left(\frac{\pi}{2}\right) > 0. \quad (36)$$

Since $\cos \pi x = -\cos \pi X$, Equation (6) and Theorem 4.1 give

$$\mathcal{D}_c(x) \sim C_* \left(\frac{X}{e}\right)^{X/2} \exp\left(-\frac{\rho}{2} \cos \pi X\right) \exp\left(\sum_{k \geq 1} c_k X^{1-2k}\right). \quad (37)$$

The periodic factor is order one and must not be hidden inside an algebraic remainder. If desired, it has the exact Fourier–Bessel expansion

$$\exp\left(-\frac{\rho}{2} \cos \theta\right) = I_0\left(\frac{\rho}{2}\right) + 2 \sum_{m \geq 1} (-1)^m I_m\left(\frac{\rho}{2}\right) \cos(m\theta). \quad (38)$$

At integral $x = n$, $\cos \pi X = -(-1)^n$, so the periodic normalization reduces exactly to C_n/C_* in Equation (24).

5 Parity-resolved functional inverses

5.1 Existence and exact Lambert core

For each δ , the branch $\mathcal{D}_\delta(x)$ is positive and eventually strictly increasing, because

$$\frac{\mathcal{D}'_\delta(x)}{\mathcal{D}_\delta(x)} = \frac{1}{2} \log 2 + \frac{1}{2} \psi\left(\frac{x}{2} + 1\right) = \frac{1}{2} \log x + O(x^{-1}). \quad (39)$$

Let $\mathcal{D}_\delta^{-1}(y)$ denote its large real inverse. The two continuations are exact scalar multiples,

$$\mathcal{D}_1(x) = \sqrt{\frac{2}{\pi}} \mathcal{D}_0(x), \quad \mathcal{D}_1^{-1}(y) = \mathcal{D}_0^{-1}\left(\sqrt{\frac{\pi}{2}} y\right), \quad (40)$$

which is the exact origin of the universal coefficient blocks below.

Put

$$R_\delta = \log\left(\frac{y}{C_\delta}\right), \quad w_\delta = W_0\left(\frac{2R_\delta}{e}\right), \quad (41)$$

$$T_\delta = \frac{2R_\delta}{w_\delta} = e^{w_\delta+1}, \quad \Lambda_\delta = \log T_\delta = w_\delta + 1. \quad (42)$$

Then

$$\Phi(T_\delta) = R_\delta, \quad T_\delta(\Lambda_\delta - 1) = 2R_\delta, \quad \Phi'(T_\delta) = \frac{\Lambda_\delta}{2}. \quad (43)$$

The principal branch W_0 is the only real Lambert branch needed because $2R_\delta/e > 0$ for large y .

For notational economy, the subscripts δ on R, w, T, Λ are suppressed in coefficient calculations below.

5.2 Normalized displacement equation

Write

$$X = T(1 + \epsilon), \quad q = T^{-1}, \quad z = q^2. \quad (44)$$

Subtracting the core equation $\Phi(T) = R$ from the logarithmic forward phase and multiplying by $2/T$ gives the exact formal normal form

$$\Lambda\epsilon + (1 + \epsilon)\log(1 + \epsilon) - \epsilon + 2 \sum_{k \geq 1} c_k z^k (1 + \epsilon)^{-(2k-1)} = 0. \quad (45)$$

Because the perturbation uses only powers $z^k = q^{2k}$, its unique formal solution has the form

$$\epsilon = E(z) = \sum_{n \geq 1} p_n(\Lambda) z^n. \quad (46)$$

Since $X - T = T\epsilon$, this produces odd powers $T^{-(2n-1)}$ in the absolute displacement.

Main result

For $\delta \in \{0, 1\}$, define $R_\delta, T_\delta, \Lambda_\delta$ by [Equations \(41\) and \(42\)](#). Then for every fixed $N \geq 0$,

$$\mathcal{D}_\delta^{-1}(y) = -1 + T_\delta + \sum_{n=1}^N \frac{p_n(\Lambda_\delta)}{T_\delta^{2n-1}} + O\left(\frac{1}{T_\delta^{2N+1}\Lambda_\delta}\right) \quad (47)$$

as $y \rightarrow +\infty$. All parity dependence is contained in $R_\delta = \log(y/C_\delta)$; the coefficient functions p_n are universal.

5.3 Triangular coefficient recurrence

Let

$$E_{n-1}(z) = \sum_{j=1}^{n-1} p_j(\Lambda) z^j. \quad (48)$$

Theorem 5.1 (All-orders inverse coefficient engine). *The coefficients in [Equation \(47\)](#) are uniquely determined by*

$$p_n(\Lambda) = -\frac{1}{\Lambda} [z^n] \left\{ (1 + E_{n-1}) \log(1 + E_{n-1}) - E_{n-1} + 2 \sum_{k=1}^n c_k z^k (1 + E_{n-1})^{-(2k-1)} \right\}. \quad (49)$$

Moreover,

$$p_n(\Lambda) \in \Lambda^{-1} \mathbb{Q}[\Lambda^{-1}], \quad \deg_{\Lambda^{-1}} p_n \leq 2n - 1, \quad (50)$$

and its two extreme coefficients are

$$p_n(\Lambda) = -\frac{2c_n}{\Lambda} + O(\Lambda^{-2}), \quad [\Lambda^{-(2n-1)}] p_n = \frac{1}{3^n} \binom{1/2}{n}. \quad (51)$$

Proof. At order z^n , the only occurrence of the unknown p_n in [Equation \(45\)](#) is $\Lambda p_n z^n$. All other terms involve only $p_1, \dots, p_{n-1}$, which proves existence, uniqueness, and [Equation \(49\)](#).

The nonlinear kernel starts at $E^2/2$, and all binomial coefficients in the Bernoulli sector are rational. Inductively, the coefficient inside braces is a polynomial in Λ^{-1} of degree at most $2n - 2$; division by Λ proves [Equation \(50\)](#). Its part independent of lower coefficients is $2c_n$, proving the first statement in [Equation \(51\)](#).

For the highest possible pole $\Lambda^{-(2n-1)}$, only the quadratic kernel and the first Bernoulli coefficient can contribute. Since $2c_1 = -1/6$, the maximal-pole generating equation is

$$\Lambda E + \frac{1}{2}E^2 - \frac{1}{6}z = 0.$$

The solution with zero constant term is

$$E = -\Lambda + \sqrt{\Lambda^2 + z/3} = \sum_{n \geq 1} \binom{1/2}{n} \frac{z^n}{3^n \Lambda^{2n-1}},$$

which proves the second statement. $\square$

5.4 Explicit Bell-polynomial formula

The recurrence can be expanded into a finite combinatorial sum. Use $B_{0,0} = 1$ and $B_{r,0} = 0$ for $r > 0$.

Corollary 5.2 (Partial-Bell form). *For $n \geq 1$,*

$$p_n(\Lambda) = -\frac{1}{\Lambda} \left[\sum_{m=2}^n \frac{(-1)^m (m-2)!}{n!} B_{n,m}(1!p_1, 2!p_2, \dots) + 2 \sum_{k=1}^n c_k \sum_{m=0}^{n-k} \binom{1-2k}{m} \frac{m!}{(n-k)!} B_{n-k,m}(1!p_1, 2!p_2, \dots) \right]. \quad (52)$$

Every p_j occurring on the right has $j < n$.

Proof. Use

$$(1+E) \log(1+E) - E = \sum_{m \geq 2} \frac{(-1)^m}{m(m-1)} E^m, \quad (53)$$

expand $(1+E)^{1-2k}$ binomially, and apply Equation (15). The factors $m!/[m(m-1)] = (m-2)!$ give the first line. $\square$

5.5 First five inverse blocks

The recurrence gives

$$p_1 = \frac{1}{6\Lambda}, \quad (54)$$

$$p_2 = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}, \quad (55)$$

$$p_3 = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\Lambda^5}, \quad (56)$$

$$p_4 = -\frac{1}{5443200\Lambda^7} (822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 + 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625), \quad (57)$$

$$p_5 = \frac{1}{359251200\Lambda^9} (309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 + 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 + 812350\Lambda^2 + 242550\Lambda + 40425). \quad (58)$$

Thus, through three correction blocks,

$$\begin{aligned} \mathcal{D}_\delta^{-1}(y) = & -1 + T + \frac{1}{6T\Lambda} - \frac{14\Lambda^2 + 10\Lambda + 5}{360T^3\Lambda^3} \\ & + \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360T^5\Lambda^5} + O\left(\frac{1}{T^7\Lambda}\right). \end{aligned} \quad (59)$$

5.6 Scaling from the half-shifted inverse Gamma problem

Let

$$a_k = \frac{B_{2k}(1/2)}{2k(2k-1)}. \quad (60)$$

For the half-shifted inverse Gamma problem, the normalized equation has the same kernel as Equation (45), with perturbation coefficients a_k instead of $2c_k$. By Equation (19),

$$2c_k = 4^k a_k. \quad (61)$$

The coefficient recurrence is homogeneous with respect to the weight $\text{wt}(a_k) = k$. Therefore, if $d_{2n-1}(\Lambda)$ denotes the corresponding inverse Gamma block in the normalization of [11], then

$$p_n(\Lambda) = 4^n d_{2n-1}(\Lambda). \quad (62)$$

This identity both explains the displayed coefficients and supplies a strong independent regression test for symbolic implementations.

5.7 Analytic remainder proof

Proof of the remainder in Equation (47). Substitute $E_N(z) = \sum_{n=1}^N p_n(\Lambda) z^n$ into Equation (45). By construction, its formal residual is

$$\Lambda E_N + (1 + E_N) \log(1 + E_N) - E_N + 2 \sum_{k=1}^{N+1} c_k z^k (1 + E_N)^{-(2k-1)} = O(z^{N+1}). \quad (63)$$

Restoring the factor $T/2$ removed in the normalization turns this into a forward logarithmic residual $O(T^{-(2N+1)})$. The omitted forward Stirling terms have the same or smaller order by Equation (20). Finally,

$$\frac{d}{dX} \left(\Phi(X) + \sum_{k \geq 1} c_k X^{1-2k} \right) = \frac{1}{2} \log X + O(X^{-2}) = \frac{\Lambda}{2} + o(1) \quad (64)$$

near $X = T$. The mean-value theorem divides the phase residual by a quantity asymptotic to $\Lambda/2$, yielding $O(T^{-(2N+1)}/\Lambda)$ for the inverse error. $\square$

6 The full outer iterated-logarithmic layer

Keeping W_0 unexpanded is numerically and symbolically superior. To embed the result in a conventional polynomial-logarithmic transseries, however, set

$$Z = \frac{2R}{e}, \quad L_1 = \log Z, \quad L_2 = \log L_1. \quad (65)$$

For $Z \rightarrow +\infty$,

$$W_0(Z) \sim L_1 - L_2 + \sum_{r \geq 1} \frac{P_r(L_2)}{L_1^r}, \quad (66)$$

where the general Lambert polynomial is

$$P_r(t) = \sum_{m=1}^r (-1)^{r-m} \left[\begin{matrix} r \\ r-m+1 \end{matrix} \right] \frac{t^m}{m!}. \quad (67)$$

The first three are

$$P_1 = t, \quad P_2 = \frac{t^2 - 2t}{2}, \quad P_3 = \frac{2t^3 - 9t^2 + 6t}{6}. \quad (68)$$

This is the standard all-orders expansion of Lambert W [3, 4].

To expand $T = 2R/W_0(Z)$, define

$$u_1 = -L_2, \quad u_j = P_{j-1}(L_2) \quad (j \geq 2), \quad (69)$$

so that $W_0(Z) = L_1(1 + \sum_{j \geq 1} u_j L_1^{-j})$. Define reciprocal polynomials $h_n(L_2)$ by

$$h_0 = 1, \quad h_n = - \sum_{j=1}^n u_j h_{n-j}. \quad (70)$$

Then

$$T \sim \frac{2R}{L_1} \sum_{n \geq 0} \frac{h_n(L_2)}{L_1^n}, \quad \Lambda = 1 + L_1 - L_2 + \sum_{r \geq 1} \frac{P_r(L_2)}{L_1^r}. \quad (71)$$

For example,

$$T \sim \frac{2R}{L_1} \left[1 + \frac{L_2}{L_1} + \frac{L_2^2 - L_2}{L_1^2} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^3} + \dots \right]. \quad (72)$$

For completely mechanical outer composition, let $A(t) = 1 + \sum_{k \geq 1} a_k t^k$ and $A(t)^\alpha = \sum_{n \geq 0} b_n^{(\alpha)} t^n$. Logarithmic differentiation gives the universal power recurrence

$$b_0^{(\alpha)} = 1, \quad n b_n^{(\alpha)} = \sum_{k=1}^n ((\alpha + 1)k - n) a_k b_{n-k}^{(\alpha)}. \quad (73)$$

Apply it first to $W_0(Z)/L_1 = 1 + \sum u_k L_1^{-k}$, and then to

$$\frac{\Lambda}{L_1} = 1 + \frac{1 - L_2}{L_1} + \sum_{k \geq 2} \frac{P_{k-1}(L_2)}{L_1^k}. \quad (74)$$

Finite convolution of the resulting power coefficients expands every monomial $T^{-(2m-1)} \Lambda^{-j}$. Thus the inner blocks require no implicit composition step after Equation (73).

Proposition 6.1 (Nested transseries hierarchy). *Substitution of Equation (71) into Equation (47) gives a unique formal expansion in the ordered hierarchy*

$$\frac{R}{L_1}, \quad \frac{R P(L_2)}{L_1^2}, \quad \frac{R P(L_2)}{L_1^3}, \dots \succ 1 \succ T^{-1} \Lambda^{-j} \succ T^{-3} \Lambda^{-j} \succ \dots. \quad (75)$$

At every prescribed monomial only finitely many Bell/Stirling partitions contribute. Therefore Equations (52), (67) and (70) constitute general coefficient formulae for the fully expanded outer-and-inner transseries.

Remark 6.2. The constant shift -1 is beyond every fixed order of the outer L_1^{-1} -series because $R/L_1^N \rightarrow \infty$ for every fixed N . The first Bernoulli inverse block is smaller still: $p_1/(T\Lambda) \asymp (12R)^{-1}$. This ordering is obscured if all terms are written as a single informal series in $1/\log y$.

7 Inverse of the OEIS cosine interpolation

The periodic interpolation $\mathcal{D}_c$ has a qualitatively new inverse sector. Put

$$R_* = \log\left(\frac{y}{C_*}\right), \quad w_* = W_0\left(\frac{2R_*}{e}\right), \quad T = \frac{2R_*}{w_*}, \quad \Lambda = 1 + w_*. \quad (76)$$

Set

$$X = T + S, \quad q = T^{-1}, \quad \theta = \pi T, \quad G(u) = (1 + u) \log(1 + u) - u. \quad (77)$$

The exact formal equation obtained from Equation (37) is

$$\begin{aligned} \mathcal{F}(q, S) &= \frac{\Lambda}{2} S + \frac{G(qS)}{2q} + \sum_{k \geq 1} c_k q^{2k-1} (1 + qS)^{-(2k-1)} \\ &\quad - \frac{\rho}{2} \cos(\theta + \pi S) = 0. \end{aligned} \quad (78)$$

The quotient $G(qS)/q$ is regular at $q = 0$, because $G(u) = u^2/2 + O(u^3)$. During coefficient extraction in q , the slowly varying Λ and the rapid phase $\theta = \pi T$ are treated as coefficient parameters; this is an oscillatory logarithmic transseries rather than an ordinary power series with constant coefficients.

7.1 The periodic carrier

At $q = 0$, the displacement s_0 is determined by

$$s_0 = \mu \cos(\theta + \pi s_0), \quad \mu = \frac{\rho}{\Lambda}. \quad (79)$$

It is $O(\Lambda^{-1})$, much larger than the first Bernoulli inverse block $O(T^{-1}\Lambda^{-1})$.

Theorem 7.1 (All-orders periodic carrier). *For sufficiently large Λ , the solution of Equation (79) tending to zero has the convergent local Lagrange–Bürmann expansion*

$$s_0(\theta, \Lambda) = \sum_{m \geq 1} \frac{\mu^m}{m!} \left. \frac{d^{m-1}}{du^{m-1}} \cos^m(\theta + \pi u) \right|_{u=0}. \quad (80)$$

Equivalently,

$$s_0 = \sum_{m \geq 1} \frac{\mu^m}{m! 2^m} \sum_{j=0}^m \binom{m}{j} (i\pi(m-2j))^{m-1} e^{i(m-2j)\theta}. \quad (81)$$

The expression in Equation (81) is real. Its first terms are

$$s_0 = \mu \cos \theta - \frac{\pi}{2} \mu^2 \sin 2\theta - \frac{\pi^2}{8} \mu^3 (\cos \theta + 3 \cos 3\theta) + O(\mu^4). \quad (82)$$

Proof. Equation (79) has the form $s_0 = \mu \phi(s_0)$ with $\phi(u) = \cos(\theta + \pi u)$. Lagrange inversion gives Equation (80). Expanding

$$\cos^m z = 2^{-m} \sum_{j=0}^m \binom{m}{j} e^{i(m-2j)z}$$

and differentiating gives Equation (81). Pairing j with $m-j$ proves reality. Direct evaluation for $m = 1, 2, 3$ gives Equation (82). $\square$

7.2 All algebraic corrections around the carrier

Seek

$$S(q) = s_0 + \sum_{r \geq 1} s_r(\theta, \Lambda) q^r. \quad (83)$$

The carrier Jacobian is

$$J = \partial_S \mathcal{F}(0, s_0) = \frac{1}{2} (\Lambda + \rho \pi \sin(\theta + \pi s_0)). \quad (84)$$

It is nonzero for all sufficiently large Λ .

Theorem 7.2 (Periodic triangular recurrence). *Let $S_{r-1} = s_0 + \sum_{j=1}^{r-1} s_j q^j$. Then every coefficient in Equation (83) is uniquely determined by*

$$s_r = -\frac{1}{J} [q^r] \mathcal{F}(q, S_{r-1}), \quad r \geq 1. \quad (85)$$

In particular,

$$s_1 = \frac{1/12 - s_0^2/4}{J}. \quad (86)$$

For fixed N , the large inverse of the cosine interpolation is

$$\mathcal{D}_c^{-1}(y) = -1 + T + s_0(\theta, \Lambda) + \sum_{r=1}^N \frac{s_r(\theta, \Lambda)}{T^r} + O\left(\frac{1}{T^{N+1}\Lambda}\right). \quad (87)$$

Each s_r can in turn be expanded to arbitrary order in $\mu = \rho/\Lambda$ by Equation (80) and ordinary Bell composition.

Proof. At order q^r , the missing term $s_r q^r$ enters linearly with coefficient J ; all remaining contributions depend only on earlier coefficients. This proves Equation (85). At order q ,

$$\frac{G(qS)}{2q} = \frac{q}{4} s_0^2 + O(q^2), \quad \sum_{k \geq 1} c_k q^{2k-1} (1 + qS)^{-(2k-1)} = c_1 q + O(q^2),$$

so $J s_1 + s_0^2/4 + c_1 = 0$; using $c_1 = -1/12$ gives Equation (86). Truncation through q^N leaves a phase residual $O(q^{N+1})$. Since the exact phase derivative is $\Lambda/2 + O(1)$, the root error is $O(q^{N+1}/\Lambda)$, proving Equation (87). $\square$

Important distinction

The periodic carrier in Equation (80) is not a property of the discrete sequence alone. It records the choice of the cosine interpolation Equation (6). A different smooth interpolation agreeing at all integers can change this sector while leaving every integer value and both parity-resolved transseries unchanged.

8 Algorithms and exact symbolic computation

8.1 Forward coefficients

Algorithm 8.1 (Forward and reciprocal coefficient table). Given a truncation order M :

1. compute c_k from Equation (19) for $1 \leq k \leq \lceil M/2 \rceil$;
2. set $\lambda_{2k-1} = c_k$ and $\lambda_{2k} = 0$;
3. compute $A_n^{(\alpha)}$ either from the complete Bell formula Equation (29) or the recurrence Equation (31).

Only exact rational arithmetic is required when $\alpha \in \mathbb{Q}$.

8.2 Inverse blocks

Algorithm 8.2 (Parity-resolved inverse blocks). The following truncated-series procedure computes $p_1, \dots, p_N$ exactly in $\mathbb{Q}(\Lambda)$.

```

E := 0
for n := 1, ..., N:
  H := (1+E) log(1+E) - E
  for k := 1, ..., n:
    H := H + 2 c_k z^k (1+E)^(-(2k-1))
  p_n := - coefficient(truncate_z(H, n), z^n) / Lambda
  E := E + p_n z^n
return p_1, ..., p_N

```

At step n , every product and composition may be truncated immediately above z^n . The symbolic residual check is

$$\Lambda E_N + (1 + E_N) \log(1 + E_N) - E_N + 2 \sum_{k=1}^N c_k z^k (1 + E_N)^{-(2k-1)} = O(z^{N+1}). \quad (88)$$

A compact Wolfram Language implementation is:

```

Clear[c, p, z, lam, E];
c[k_Integer?Positive] :=
  (1 - 2^(2 k - 1)) BernoulliB[2 k]/(2 k (2 k - 1));

inverseBlocks[N_Integer?NonNegative] := Module[{E = 0, H, out = {}},
  Do[
    H = (1 + E) Log[1 + E] - E
      + Sum[2 c[k] z^k (1 + E)^(-(2 k - 1)), {k, 1, n}];
    p[n] = -Coefficient[Normal@Series[H, {z, 0, n}], z, n]/lam;
    p[n] = Factor[p[n]];
    E += p[n] z^n;
    AppendTo[out, p[n]],
    {n, 1, N}];
  out
]

```

8.3 Stable evaluation

For a target supplied as $Y = \log y$, never form y when it would overflow. For branch δ , compute

$$R = Y - \log C_\delta, \quad w = W_0(2R/e), \quad T = 2R/w, \quad \Lambda = w + 1, \quad (89)$$

then evaluate [Equation \(47\)](#). A final Newton correction in the logarithmic phase is

$$x_{\text{new}} = x - \frac{\log \mathcal{D}_\delta(x) - Y}{\frac{1}{2} \log 2 + \frac{1}{2} \psi(x/2 + 1)}. \quad (90)$$

This avoids overflow and is quadratically convergent once the asymptotic approximation is in the local basin.

9 Numerical verification

All values in this section were computed with 80-digit arithmetic from [Equation \(2\)](#); no rounded forward values of y were used.

9.1 Forward series

Let $F_M(x)$ denote [Equation \(32\)](#) truncated after $A_M^{(1)} X^{-M}$. The table gives $|F_M(x)/\mathcal{D}_\delta(x) - 1|$, with $\delta \equiv x \pmod{2}$.

x	parity	$M = 0$	$M = 1$	$M = 3$	$M = 5$	$M = 9$
10	even	$7.590 \cdot 10^{-3}$	$4.330 \cdot 10^{-5}$	$2.599 \cdot 10^{-7}$	$4.974 \cdot 10^{-9}$	$1.388 \cdot 10^{-11}$
11	odd	$6.957 \cdot 10^{-3}$	$3.538 \cdot 10^{-5}$	$1.751 \cdot 10^{-7}$	$2.782 \cdot 10^{-9}$	$5.454 \cdot 10^{-12}$
20	even	$3.974 \cdot 10^{-3}$	$9.988 \cdot 10^{-6}$	$1.432 \cdot 10^{-8}$	$6.756 \cdot 10^{-11}$	$1.314 \cdot 10^{-14}$
21	odd	$3.793 \cdot 10^{-3}$	$9.014 \cdot 10^{-6}$	$1.166 \cdot 10^{-8}$	$4.972 \cdot 10^{-11}$	$7.956 \cdot 10^{-15}$
50	even	$1.635 \cdot 10^{-3}$	$1.483 \cdot 10^{-6}$	$3.106 \cdot 10^{-10}$	$2.113 \cdot 10^{-13}$	$9.594 \cdot 10^{-19}$
51	odd	$1.604 \cdot 10^{-3}$	$1.424 \cdot 10^{-6}$	$2.861 \cdot 10^{-10}$	$1.866 \cdot 10^{-13}$	$7.799 \cdot 10^{-19}$

Table 1: Relative errors of the centered multiplicative expansion.

9.2 Inverse series

Let $y = \mathcal{D}_\delta(x)$ exactly and let $I_N(y)$ denote Equation (47) through p_N/T^{2N-1} ; $I_0 = -1 + T$. The following table gives $|I_N(y) - x|$.

x	parity	$N = 0$	$N = 1$	$N = 2$	$N = 3$	$N = 5$
10	even	$6.307 \cdot 10^{-3}$	$1.644 \cdot 10^{-5}$	$1.610 \cdot 10^{-7}$	$3.689 \cdot 10^{-9}$	$1.139 \cdot 10^{-11}$
11	odd	$5.581 \cdot 10^{-3}$	$1.210 \cdot 10^{-5}$	$9.981 \cdot 10^{-8}$	$1.934 \cdot 10^{-9}$	$4.263 \cdot 10^{-12}$
20	even	$2.605 \cdot 10^{-3}$	$1.752 \cdot 10^{-6}$	$4.775 \cdot 10^{-9}$	$3.103 \cdot 10^{-11}$	$7.651 \cdot 10^{-15}$
21	odd	$2.450 \cdot 10^{-3}$	$1.496 \cdot 10^{-6}$	$3.717 \cdot 10^{-9}$	$2.204 \cdot 10^{-11}$	$4.523 \cdot 10^{-15}$
50	even	$8.311 \cdot 10^{-4}$	$8.979 \cdot 10^{-8}$	$4.194 \cdot 10^{-11}$	$4.716 \cdot 10^{-14}$	$3.443 \cdot 10^{-19}$
51	odd	$8.111 \cdot 10^{-4}$	$8.422 \cdot 10^{-8}$	$3.785 \cdot 10^{-11}$	$4.094 \cdot 10^{-14}$	$2.767 \cdot 10^{-19}$
100	even	$3.575 \cdot 10^{-4}$	$9.580 \cdot 10^{-9}$	$1.147 \cdot 10^{-12}$	$3.316 \cdot 10^{-16}$	$1.590 \cdot 10^{-22}$

Table 2: Absolute errors of the parity-resolved inverse transseries.

The near-geometric improvement at these moderate arguments is not a claim of convergence. At fixed x , the Bernoulli-driven expansion eventually diverges; optimal truncation or a Newton correction should be used in high precision work.

10 Beyond all algebraic orders

The coefficients c_k inherit the factorial growth of Bernoulli numbers. Since

$$B_{2k} \sim (-1)^{k-1} \frac{2(2k)!}{(2\pi)^{2k}}, \quad (91)$$

the least term of the centered series occurs for $2k$ of order πX , and the optimally truncated positive-real remainder is exponentially small on the scale $e^{-\pi X}$, up to algebraic factors. Exponentially improved Gamma expansions express this remainder by terminant functions [6].

The inverse transport is universal. Suppose an exponentially improved forward logarithmic phase is written

$$\log \mathcal{D}_\delta(x) = \log C_\delta + \Phi(X) + H_{\text{alg}}(X) + \mathcal{E}(X), \quad (92)$$

where $\mathcal{E}$ is beyond every algebraic power. If X_{alg} is the algebraic inverse transseries, then the first beyond-all-orders inverse correction is

$$\Delta X = -\frac{\mathcal{E}(X_{\text{alg}})}{\Phi'(X_{\text{alg}}) + H'_{\text{alg}}(X_{\text{alg}})} + O(\mathcal{E}^2) \sim -\frac{2\mathcal{E}(T)}{\Lambda}. \quad (93)$$

Higher powers of $\mathcal{E}$ follow from ordinary Lagrange–Bürmann reversion in the phase coordinate. Thus an optimally truncated forward sector of scale $e^{-\pi T}$ induces an inverse sector of scale $e^{-\pi T}/\Lambda$. This section supplies the bridge to a resurgent or hyperasymptotic completion without pretending that the divergent Bernoulli series is convergent.

11 Logical status and interpretation

Three kinds of statement have been used.

Exact identities. Equations [Equations \(2\)](#), [\(4\)](#) and [\(6\)](#) are exact wherever the gamma factors are defined. The Lambert identities [Equation \(43\)](#) are also exact.

Formal coefficient identities. The Bell formulae and triangular recurrences are identities in formal power series over $\mathbb{Q}(\Lambda)$, or over the ring of periodic functions of θ and formal Λ^{-1} -series in the cosine case. Their coefficient uniqueness is algebraic.

Poincaré asymptotics. The forward remainder follows from standard positive-real Stirling estimates. Eventual monotonicity and the residual-to-root argument prove the stated fixed-order inverse remainders. No convergence of the infinite Bernoulli series is asserted.

12 Conclusions

The double factorial has a particularly clean all-orders structure after the single shift $X = x + 1$:

1. The even and odd branches share the same centered logarithmic tail; parity survives only in $C_0 = \sqrt{\pi}$ and $C_1 = \sqrt{2}$.
2. The forward coefficients are explicit Bernoulli combinations $c_k = (1 - 2^{2k-1})B_{2k}/[2k(2k-1)]$, and every multiplicative, reciprocal, or fixed-power coefficient is a complete Bell polynomial.
3. The large inverse is controlled exactly by $T = 2R/W_0(2R/e)$. Its correction support is the odd grid $T^{-1}, T^{-3}, \dots$, and the universal blocks $p_n(\Lambda)$ are generated by the finite recurrence [Equation \(49\)](#) or the Bell formula [Equation \(52\)](#).
4. The complete outer logarithmic layer follows from the unsigned-Stirling formula [Equation \(67\)](#) and the reciprocal recurrence [Equation \(70\)](#).
5. Merging parity classes by the OEIS cosine interpolation creates a new periodic carrier of order $1/\Lambda$. Its coefficients are given by the Lagrange/Fourier formula [Equation \(81\)](#); all subsequent powers of T^{-1} follow from [Equation \(85\)](#).

Summary

For symbolic work, centre at $X = x + 1$, retain Lambert W_0 exactly, generate c_k from Bernoulli numbers, exponentiate with complete Bell polynomials, and invert with the triangular $z = T^{-2}$ recurrence. Expand W_0 into iterated logarithms only as a final presentation layer. When parity is not specified, state the interpolation explicitly: its choice changes the inverse transseries.

A A hand derivation of the first two inverse blocks

Since

$$(1 + E) \log(1 + E) - E = \frac{1}{2}E^2 - \frac{1}{6}E^3 + \dots, \quad (94)$$

the coefficient of z in [Equation \(45\)](#) is

$$\Lambda p_1 + 2c_1 = 0,$$

whence $p_1 = 1/(6\Lambda)$. At order z^2 ,

$$\Lambda p_2 + \frac{1}{2}p_1^2 - 2c_1p_1 + 2c_2 = 0. \quad (95)$$

Using $c_1 = -1/12$, $c_2 = 7/360$, and $p_1 = 1/(6\Lambda)$ gives

$$p_2 = -\frac{1}{\Lambda} \left(\frac{7}{180} + \frac{1}{36\Lambda} + \frac{1}{72\Lambda^2} \right) = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}.$$

This illustrates why the recurrence produces a finite polynomial in Λ^{-1} at every order.

B A sixth inverse block

For regression testing at higher order, the next coefficient is

$$\begin{aligned} p_6(\Lambda) = -\frac{1}{5884534656000\Lambda^{11}} & (46195687238400\Lambda^{10} + 8854370366400\Lambda^9 + 2171643318000\Lambda^8 \\ & + 605515022112\Lambda^7 + 212869408752\Lambda^6 + 84136852800\Lambda^5 \\ & + 35589153600\Lambda^4 + 14234019800\Lambda^3 + 4868113250\Lambda^2 \\ & + 1213962750\Lambda + 165540375). \end{aligned} \quad (96)$$

Its leading coefficient is $-2c_6/\Lambda$, and its highest pole equals $3^{-6} \binom{1/2}{6} \Lambda^{-11}$, as required by Equation (51).

C Notation dictionary

Symbol	Meaning
δ	Parity label: 0 for the even branch, 1 for the odd branch.
$\mathcal{D}_\delta$	Exact gamma continuation of one parity subsequence.
$\mathcal{D}_c$	OEIS cosine interpolation agreeing with both parity classes.
$X = x + 1$	Optimal centred variable.
C_0, C_1	Branch constants $\sqrt{\pi}, \sqrt{2}$.
C_*	Mean cosine-interpolation constant $(2\pi)^{1/4}$.
ρ	Periodic amplitude parameter $\frac{1}{2} \log(\pi/2)$.
$\Phi(X)$	Dominant phase $X(\log X - 1)/2$.
c_k	Centered logarithmic Stirling coefficient in Equation (19).
λ_r	Sparse embedding of c_k on odd indices.
$A_n^{(\alpha)}$	Multiplicative coefficient for the power α .
R	Logarithmic target after removal of the branch constant.
w	Lambert variable $W_0(2R/e)$.
T	Exact Lambert core $2R/w = e^{w+1}$.
Λ	Core slope parameter $\log T = w + 1$.
q, z	Inner grid variables $q = T^{-1}$, $z = q^2$.
$p_n(\Lambda)$	Universal parity-resolved inverse block multiplying $T^{-(2n-1)}$.
L_1, L_2	Outer logarithms $\log(2R/e)$, $\log L_1$.
P_r	Lambert polynomial in Equation (67).
s_0	Periodic carrier for the cosine interpolation inverse.
s_r	Algebraic correction around the periodic carrier.

References

- [1] OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, A006882, *Double factorials*: $n!! = 1 \cdot 3 \cdot 5 \cdots n$ or $2 \cdot 4 \cdot 6 \cdots n$, <https://oeis.org/A006882>, accessed 3 September 2026.

- [2] B. E. Meserve, Double factorials, *American Mathematical Monthly* **55** (1948), 425–426.
- [3] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359, <https://doi.org/10.1007/BF02124750>.
- [4] NIST Digital Library of Mathematical Functions, §4.13, Lambert W -function, <https://dlmf.nist.gov/4.13>, accessed 3 September 2026.
- [5] NIST Digital Library of Mathematical Functions, §5.11, Asymptotic expansions of the Gamma function, <https://dlmf.nist.gov/5.11>, accessed 3 September 2026.
- [6] G. Nemes, Error bounds and exponential improvements for the asymptotic expansions of the Gamma function and its reciprocal, *Proceedings of the Royal Society of Edinburgh Section A: Mathematics* **145** (2015), 571–596, <https://doi.org/10.1017/S0308210513001558>.
- [7] F. W. J. Olver, *Asymptotics and Special Functions*, A K Peters, Wellesley, MA, 1997 reprint of the 1974 edition.
- [8] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics 1888, Springer, Berlin, 2006.
- [9] V. Reshetnikov, *Combinatorial Coefficient Calculus: A Unified Formulae Monograph*, ProveIt repository, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial_Coefficient_Calculus, accessed 3 September 2026.
- [10] V. Reshetnikov, *Series and Transseries*, ProveIt research-frontier notes, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>, accessed 3 September 2026.
- [11] V. Reshetnikov, *Asymptotic Inversion of the Gamma and Barnes G -Functions*, ProveIt repository, 2026, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries/special-function-inversion/Asymptotic_Inversion_Gamma_Barnes_G.